GITAM Deemed to be University

Artificial Intelligence and Machine Learning | Course Code: AI201

Artificial Intelligence & Search Strategies - Laboratory Manual & Practi

---

Syllabus Focus: AI201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Intelligent Agents & State Space Search

" Syllabus Scope: Agent architectures, PEAS definitions, Breadth-First, Uniform Cost, Depth-First, and It

Unit 2: Informed Heuristic Search & Optimality

" Syllabus Scope: Greedy best-first search, A* search algorithm, Admissible and consistent heuristics, Me

AI201: Artificial Intelligence & Search Strategies - Advanced Topics & Assessment

Academic Track: Artificial Intelligence and Machine Learning - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Adversarial Search & Games

" Game theory formulation, Minimax decision algorithm, Alpha-Beta pruning optimality, Evaluation functions for

Unit 4: Constraint Satisfaction Problems CSP

" Constraint propagation, AC-3 algorithm, Backtracking with Minimum Remaining Values MRV heuristic, Forward ch

Unit 5: Knowledge Representation & Logic

" Propositional logic, First-order predicate logic, Inference rules, Forward and backward chaining, Resolution

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Uninformed Search BFS, DFS, Uniform Cost for Pathfinding in a Grid
Lab 2: Implementation of A* Heuristic Search for the 8-Puzzle Problem using Manhattan Distance
Lab 3: Building a Tic-Tac-Toe Game Engine using Adversarial Minimax with Alpha-Beta Pruning
Lab 4: Constraint Satisfaction Solver for Map Coloring and Sudoku using AC-3 Arc Consistency

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Stuart Russell, Peter Norvig - Artificial Intelligence: A Modern Approach Pearson 4th Ed
* Elaine Rich, Kevin Knight - Artificial Intelligence McGraw-Hill

Authorized NoteVault Educational Document " Verified for Registered University Students